Nathan David Obeng-Amoako, E.I.T.

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RESEARCH INTERESTS

- Intelligent Transportation Systems
- Advanced Traffic Signal Control
- Machine Learning and Deep Learning
- Transportation Demand Forecasting
- Transit Operations and Management
- Bicycle and Pedestrian Transportation

EDUCATION

Doctor of Philosophy in Civil Engineering (Transportation) **Sep 2023 – Present**
Northeastern University, Boston, MA (GPA: 4.00/4.00)
Research Advisor: Professor Peter Furth

Master of Science in Civil Engineering (Transportation) **Sep 2023 – Apr 2025**
Northeastern University, Boston, MA (GPA: 4.00/4.00)

Bachelor of Science in Civil Engineering **Aug 2018 – Sep 2022**
Kwame Nkrumah University of Science and Technology (KNUST), Kumasi
Recognition: Best Graduating Student, Class of 2022

RESEARCH EXPERIENCE

Graduate Research Assistant, Northeastern University, Boston, MA **Sep 2023 – Present**
- Development of advanced traffic signal controls and simulations
- Investigation of barriers to low-stress bicycle network connectivity

Research Assistant, Civil Engineering Department, KNUST **Jan 2022 – Sep 2022**
- Empirical investigation of aggregate retention benefits of waste-modified asphalt
- Analysis of mechanical and durability properties of asphalt mixtures

SPONSORED RESEARCH PROJECTS

Project: Bicycle Level of Traffic Stress at Intersections (xLTS) **Jun 2024 – Aug 2025**
Funding Agency: United States Department of Transportation (USDOT)
- Assessing the impact of high-stress crossings on bike network connectivity
- Developing algorithms and data structures for traffic stress network analysis

Project: Accessible Bus Stop Design in the Presence of Bike Lanes **Sep 2023 – Jun 2024**
Funding Agency: Massachusetts Department of Transportation (MassDOT)
- Studied bus rider and bicyclist behavior at floating and constrained bus stops
- Proposed design improvements to reduce bus-rider-and-bicyclist conflicts

JOURNAL PUBLICATIONS

2025

1. Peter Furth, Ray Saeidi-Razavi, **Nathan David Obeng-Amoako** and Milad Tahmasebi**.** TSP-Friendly Underlying Traffic Signal Control: An Essential Complement of Transit Signal Priority. *Future Transportation* (2025), doi: https://doi.org/10.3390/futuretransp5040160

2. Joshua Kofi Asamoah, Blessing Agyei Kyem, **Nathan David Obeng-Amoako**, Armstrong Aboah, SAAM-ReflectNet: Sign-Aware Attention-Based Multitasking Framework for Integrated Traffic Sign Detection and Retroreflectivity Estimation, *Expert Systems With Applications* (2025), doi: https://doi.org/10.1016/j.eswa.2025.128003

Submitted and under review:

2026

3. Yu-Min (Thomas) Yanga, Dewan Tanvir Ahammed, **Nathan David Obeng-Amoako**, Chengbo Ai, Peter Furth and Eleni Christofa. Community-Engaged Informed Floating Bus Stop Guidelines. *Transportation Research Record*.

CONFERENCE PRESENTATIONS

2026

1. Yu-Min (Thomas) Yanga, Dewan Tanvir Ahammed, **Nathan David Obeng-Amoako**, Eleni Christofa, Chengbo Ai, and Peter Furth Community-Engaged Informed Floating Bus Stop Guidelines. *Transportation Research Board Annual Meeting 2026*, Washington, D.C.

2025

2. **Nathan David Obeng-Amoako** and Peter Furth. (May 2024). Underlying Traffic Signal Control: The Key to Effective Transit Signal Priority – A Case Study of MBTA Bus Route 39 along South Huntington Avenue, Boston, MA. *2025 MassDOT Transportation Innovation Conference*, Worcester, MA.

3. Yu-Min (Thomas) Yanga, Dewan Tanvir Ahammed, **Nathan David Obeng-Amoako**, Eleni Christofa, Chengbo Ai, and Peter Furth. Design Guidelines for Accessible Floating Bus Stops. *International Conference on Transportation and Development 2025, ASCE*, Glendale, AZ.

2024

4. Yu-Min (Thomas) Yanga, Dewan Tanvir Ahammed, **Nathan David Obeng-Amoako**, Eleni Christofa, Chengbo Ai, and Peter Furth. (April 2024). Accessible Bus Stop Design in the Presence of Bike Lanes. Poster. *2024 MassDOT Transportation Innovation Conference*, Worcester, MA.

2023

5. **Nathan David Obeng-Amoako**, Arthur Louis Senaya, Bundu Kassim, Francisca Owusu-Ansah, Eugene Damoah, Santus Worclachie, and Kenneth A. Tutu. (April 2023). Modification of Bitumen with Waste Materials for Enhanced Aggregate Retention in Surface-Dressed Roads. Lectern. *3rd IRF Africa Regional Congress & Exhibition*, Accra, Ghana.

RESEARCH REPORTS

2024

1. Eleni Christofa, Chengbo Ai, Peter Furth, Yu-Min Yang, Dewan Tanvir Ahammed, and **Nathan David Obeng-Amoako**. (2024). *Accessible Bus Stop Design in the Presence of Bike Lanes*. Massachusetts Department of Transportation (MassDOT), Office of Transportation Planning. Report No. 24-060. Retrieved from https://rosap.ntl.bts.gov/view/dot/79327

JOURNAL REVIEWS

1. Transportation Research Record, SAGE Publications **2024 – Present**
2. Transportation Research Interdisciplinary Perspectives, Elsevier **2025 – Present**
3. Expert Systems with Applications, Elsevier **2025 – Present**

NOTABLE DISTINCTIONS AND ACHIEVEMENTS

- Grant Recipient, New England African Business Expo, Worcester, MA **May 2025**
- Civil and Environmental Engineering Distinguished Fellowship Award, Northeastern Univ **Aug 2023**
- Associated Consultants Ltd. Award for Best Graduating Student, KNUST **Jul 2023**
- Provost Award for Best & Excellent Students in College of Engineering, KNUST (x4) **2019 – 2022**
- Academic Excellence Award, Ghana Engineering Students' Association, Ghana **Sep 2022**
- Global Finalist, Unilever Future Leaders League Business Pitch Competition, U.K. **Jul 2022**
- Scholar, Tullow Oil Scholarship Scheme, Ghana **Apr 2019**
- Semifinalist, National Science and Maths Quiz (NSMQ), Ghana **Jul 2018**

RELEVANT SOFTWARE AND SKILLS

- **Python, R, MATLAB** machine learning, deep learning, advanced data analysis
- **PTV Vissim/Vistro** traffic flow simulations, traffic signal analysis, traffic impact analysis
- **Synchro** traffic flow modeling, traffic signal timing optimization
- **ArcGIS** Spatial data mapping and analysis for informed decision-making
- **Gephi** Network data analysis and visualization
- **AutoCAD** precision drafting and technical drawings, 2D and 3D
- **HTML/CSS/JavaScript** developing web applications
- **Minitab, MS Excel** data visualization, statistical computing, data organization

APPOINTMENTS AND ADVISORY ROLES

Member, Massachusetts Bicycle and Pedestrian Advisory Board, MassDOT **Jan 2026 – Present**

- Appointed by Governor Maura T. Healey to advise on state-wide bicycle and pedestrian infrastructure policies and programs
- Provide guidance on transportation infrastructure, safety, and accessibility initiatives

LEADERSHIP EXPERIENCE

Secretary, Ghana Transportation Professionals Forum (GTPF) **Feb 2026 – Present**

- Maintain membership records and coordinate executive board communications
- Support event logistics and engagements with Ghana transportation stakeholders

Committee Member, Career Mixer Committee, CEE Dep't, Northeastern Univ. **Sep 2025 – Feb 2026**

- Organizing networking events connecting students with industry
- Coordinating logistics for career dev't workshops and employer engagements

Executive C'ttee Member, CEE Graduate Student Council, Northeastern Univ. **Feb 2024 – Present**

- Brainstorming and implementing strategies to enhance graduate student experience
- Serving as a liaison between the departmental chair and the graduate student body

Department Rep., College of Engineering. PhD Council, Northeastern Univ. **Jun 2024 – Aug 2025**

- Meeting with advisor to discuss novel ways of improving student experience
- Organizing weekly coffee hours and semesterly happy hours for PhD students

Communications Director, Young Professionals and Youth Coalition, Ghana **Jan 2021 – Dec 2021**

- Social media management to curate consistent communications content
- Organization of in-person and virtual engagements with members and partners

President, Civil Engineering Students' Association, KNUST **Feb 2020 – Sep 2021**

- Coordination of fundraising events to secure administrative equipment
- Establishment of academic support scheme for underachieving students

TEACHING EXPERIENCE

Northeastern University, Boston, MA

Summers of 2024, 2025, 2026

Program Assistant, Sustainable Transportation Summer School Program

Location: Technische Universiteit Delft, The Netherlands

- Facilitate logistical, academic and evaluative support for an intensive 5-week program
- Assist in teaching Vision Zero safety principles to 25 students
- Explore strategies for car-free and car-lite zones to enhance active transportation

Spring 2026

Graduate Teaching Assistant, Energy Systems: Science, Technology, and Sustainability

- Provided grading support for coursework on energy technologies and sustainability
- Led discussions on the principles of energy, work, and thermodynamics

Spring 2026

Graduate Teaching Assistant, Transportation Systems Analysis and Planning

- Supported students during office hours with questions on design software (Synchro)
- Evaluated assignments on intersection analysis, signal timing, and street design principles

Fall 2025

Graduate Teaching Assistant, Highway Design

- Provided grading support for homework assignments
- Assisted students during office hours, addressing questions about design principles

Fall 2025

Graduate Teaching Assistant, Transportation Systems Analysis and Planning

- Evaluated coursework and provided constructive feedback on analytical assignments
- Supported students' understanding of transportation system planning frameworks

Fall 2023

Course Instructor, Recitation for Statics and Solid Mechanics

- Tutored fundamental concepts of Newtonian physics and solid mechanics
- Evaluated students' test performance and offered comprehensive feedback

Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Spring 2022 **Teaching Assistant, Irrigation & Drainage Engineering**

- Activity-oriented instruction of drainage design using *Autodesk Civil 3D®*
- Development of course outlines and preparation of lecture materials

PROFESSIONAL AFFILIATIONS

- American Society of Civil Engineers (ASCE) **Feb 2025 – Present**
- Boston Society of Civil Engineers Section (BSCES) of ASCE **Feb 2025 – Present**
- New England Intelligent Transportation Society (NEITS) **Sep 2023 – Present**
- Institute of Transportation Engineers (ITE) **Jan 2024 – Present**

VOLUNTEER EXPERIENCE & COMMUNITY INVOLVEMENT

- **Judge**, Massachusetts Science and Engineering Fair, Worcester (April 2025)
- **Cycling Activist**, Boston Cycling Union, Boston, MA (Feb 2025 – Present)
- **Executive Vice President**, Student Energy Society, KNUST (Jan 2022 – Oct 2022)
- **Community Outreach Coordinator**, Nat'l Society of Black Engineers, KNUST (Jan 2022 – Sep 2022)
- **Project Coordinator**, Engineers Without Borders, KNUST (Aug 2021 - Jul 2022)

BLOG POSTS

1. "Replacing the Reid Overpass with an At-Grade Intersection (Cambridge, near BU Bridge)." Co-authored with Peter G. Furth. January 2026. https://peterfurth.sites.northeastern.edu/2026/01/06/replacing-the-reid-overpass-with-an-at-grade-intersection-cambridge-near-bu-bridge/

2. "PhD Student Reflects on the Sustainable Transportation Dialogue of Civilizations in Holland." Northeastern University College of Engineering News, August 2024. https://coe.northeastern.edu/news/a-reflection-on-the-sustainable-transportation-dialogue-of-civilizations-course-in-holland/

3. "Bus-Friendly Traffic Signals Can Reduce Bus Delay By 90% on South Huntington Avenue." Co-authored with Peter G. Furth. April 2024. https://peterfurth.sites.northeastern.edu/2024/06/04/bus-friendly-traffic-signals-can-reduce-bus-delay-by-90-on-south-huntington-avenue/

REFERENCES

Available upon request